

Shashwat Gupta

MICROSOFT RESEARCH, COMPUTER SCIENCE @ IIT KANPUR

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(Items in purple indicate hyperlinks)

Research Interests

Agentic AI, Reliable and Safe AI, RL Post-Training for LLMs, Diffusion Models, Computer Vision for Medical Imaging, Natural Language Processing

Education

Bachelors of Technology in Computer Science (Major), Cognitive Sciences (Minor), Indian Institute of Technology (IIT) Kanpur, India, CGPA: 9.3/10 Nov. 2020 - June 2024

Exchange Semester in Computer Science, École Polytechnique Fédérale de Lausanne (EPFL), Switzerland, CGPA: 5.18/6 Jan. 2023 - July 2023

Intermediate (ISC - Class 12th), St. Fidelis College, India, 94% 2019

High School (ICSE - Class 10th), St. Fidelis College, India, 96.8% 2017

Professional Experience

Microsoft Research, India (MSRI) Bangalore, India
RESEARCH FELLOW Jan. 2025 - Present

- Architecting **AutoToolStub** in **FLASH** and **Self-Improving** code paths, a framework enhancing agentic reliability by generating pre-verified 'tool stubs' from TSGs with provenance tracking and UI-editable placeholders, establishing a foundation for trustworthy coding agents and reducing incident resolution time.
- Improved Collaborative-Problem-Solving and MultiHop-Question-Answering accuracy by **6.7%** points via optimal message structure learning, yielding a US Patent and ARR publication (**OPTIMACS**).

PwC US (Bangalore AC) Bangalore, India
GENERATIVE AI ASSOCIATE Aug. 2024 - Dec. 2024

- Engineered **end-to-end** agentic workflow for Customer Links App (CLAI) by integrating RAG and LLM capabilities.
- Reduced query latency by 30% and improved confidence by 25% using **PySpark** on a 60M-feature data fabric.

HP Tech Ventures California, US (Remote)
INVESTMENT ASSOCIATE (INTERNSHIP) June 2024 - Aug. 2024

- Evaluated **20+ AI startups** on **ARR** and **retention** metrics to identify investment targets matching the firm's thesis.
- Presented investment briefs for **4 high-potential companies** to US investors, highlighting product value and scalability.

Awards and Honours

- Awarded Dr. Prateek Mishra Memorial Scholarship for highest GPA in Electrical Engineering Dept. (2022)
- Nominated for Exchange Semester @ EPFL Switzerland (Selected top 3 among 1200 students) (2022)
- Awarded Academic Excellence Award twice for being in top 10% of batch (2021, 2022)
- Awarded FIITJEE Scholarship for All India Rank 603 (Top 0.3%) in JEE Advanced (2020)
- Secured All India Rank 1710 (Top 0.1%) in JEE Mains among 1.5 Million applicants (2020)

Publications

- OPTIMACS: Learning Optimal Message Representations for Agentic Communication (2026)**
Shashwat Gupta, Anson Bastos, Mayukh Das, Supriyo Ghosh, Nagarajan Natarajan, Chetan Bansal, Saravan Rajmohan
[Microsoft's Machine Learning, AI & Data Science Conference \(MLADS\)](#), [US Patent, Under Review in ACL Rolling Reviews](#)
- EXACT-CT: EXplainable Analysis for Crohn's and Tuberculosis using CT (2026)**
Shashwat Gupta, Sarthak Gupta, Akshan Agrawal, Mahim Naaz, Rajanikanth Yadav, Priyanka Bagade
[Under Review in RSNA Radiology Journal](#)
- FLASH: A Workflow Automation Agent for Diagnosing Recurring Incidents (2026)**
Prathamesh Deshpande, [Shashwat Gupta](#), Tanish Mittal, Xuchao Zhang, Chetan Bansal, Rujia Wang, Minghua Ma, Zhixin Ren, Hao Huang, Saravan Rajmohan
[Under Review at FSE \(ACM International Conference on the Foundations of Software Engineering\)](#)

- **KAN-Mamba FusionNet: Redefining Medical Image Segmentation with Non-Linear Modeling (2026)**
Akansh Agrawal, Akshan Agrawal, Shashwat Gupta, Priyanka Bagade
[Under Review at International Journal of Computer Vision \(IJCV\)](#)
- **CISLR: Corpus for Indian Sign Language Recognition (2023)**
Abhinav Joshi, Ashwani Bhat, Pradeep S, Priya Gole, Shreyansh Agarwal, Shashwat Gupta, Ashutosh Modi
[Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing \(EMNLP\)](#)

Teaching and Mentoring

- Teaching Assistant for ESC101A (Elementary Programming) & ESC101B (Advanced Programming) IIT Kanpur
- Invited for a [Summer Workshop on Active Learning](#) organised by Google Developer Student Club, IIT Kanpur
- Mentored 30+ Freshers on Academics and college Life as part of Counselling Service, IIT Kanpur

Academic Research Experiences

Summer Undergraduate Intern @ Exploration Lab, IIT Kanpur

IIT Kanpur

ADVISOR: PROF. ASHUTOSH MODI

May 2022 - July 2022

- Spearheaded the vision pipeline for **CISLR**, creating the largest continuous Indian Sign Language dataset (4700 words) using **OpenFace** and **OpenPose** for precise landmark extraction.
- Implemented an **I3D backbone** to handle spatiotemporal complexities, establishing a baseline with Top-1 accuracy of **3.8%** in a challenging low-resource setting.
- Published findings at **EMNLP 2022**, advancing the field of ISL processing with a publicly released dataset and robust baseline.

International Exchange Student @ CV Lab, EPFL

EPFL, Switzerland

ADVISOR: PROF. MATHIEU SALZMANN

Feb. 2023 - June 2023

- Implemented 2 novel Diffusion Architectures to reproduce the functionality of STN Modules using Diffusion Models.
- Achieved **50% accuracy** on MNIST classification from a random baseline of 10%, demonstrating the efficacy of the proposed generative approach for low-level vision tasks.

Junior Research Fellow @ IoT Lab, IIT Kanpur

IIT Kanpur

ADVISOR: PROF. PRIYANKA BAGADE

April 2024 - Oct. 2024

- Developed **EXACT-CT** to distinguish Crohn's Disease from Intestinal TB using 3D CTE scans and radiologist-identified biomarkers (e.g., necrosis, comb sign), addressing critical treatment mismanagement risks. Achieved **75% accuracy** ($p = 0.03$) by feeding mathematically derived features into an XGBoost classifier, validated via SHAP analysis and outperforming SOTA baselines like ResNet and CTFoundation.
- Proposed the theory for **KAN-Mamba FusionNet**, integrating Kolmogorov-Arnold Networks with State Space Models, achieving **3.5% improvement** over SOTA on BUSI and Kvasir-Seg datasets for medical image segmentation.

Technical Skills

Languages	Python (with Pytorch), C/C++, Assembly (Familiar),
Relevant	Machine Learning, Probability & Statistics, Data Structures and Algorithms, Convex Optimisation, Discrete Mathematics,
Courses	Linear Algebra, Differential Equations, Software Development, Numerical Methods,

Positions of Responsibility

ACM SIGCHI Student Chapter, IIT Kanpur

Kanpur, India

FOUNDER AND CHAIR

Sept. 2023 - July 2024

- Founded and established the ACM Chapter at IIT Kanpur to promote design thinking in Development.
- Supervised key projects: **JalJeevan Mission** (ML-based water leakage detection), **Saathi** (education for underprivileged children), and **Samvaad Saathi** (regional language chatbot).
- Organized **AstroDesign Odyssey** with Techkriti (50+ colleges) and hosted awareness sessions to promote HCI education.

HealthPredict (Pro-bono Initiative)

Kanpur, India

FOUNDER

Sept. 2023 - Dec. 2024

- My own pro-bono initiative post-Covid to assist hospitals and government in using ML based methods and techniques for preventive medicine and diagnosis.

Contact

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Google Scholar		Shashwat Gupta